Cambridge
O Level

**Cambridge International Examinations**
Cambridge Ordinary Level

**BIOLOGY** **5090/12**

Paper 1 Multiple Choice **October/November 2014**

**1 hour**

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

*3305766201*

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.
DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

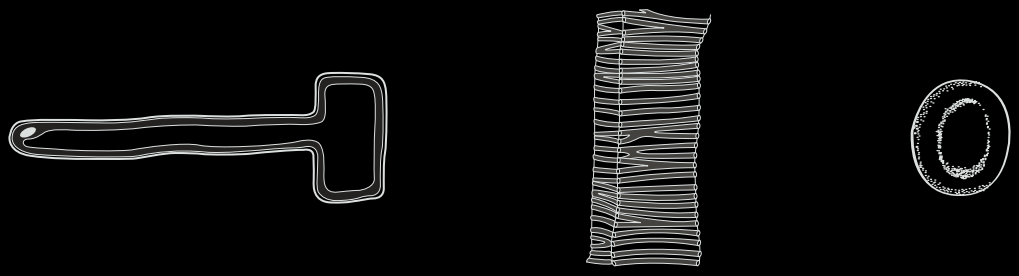
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Electronic calculators may be used.

This document consists of **18** printed pages and **2** blank pages.


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**1** The diagram shows three cellular structures.



1 2 3

Which statements about these cells are correct?

| | 1 | 2 | 3 |
|---|---|---|---|
| **A** | adapted to carry oxygen | lacks a nucleus | adapted to carry water |
| **B** | adapted to carry water | lacks a nucleus | is in contact with the soil |
| **C** | is in contact with the soil | adapted to carry water | adapted to carry oxygen |
| **D** | is in contact with the soil | is in contact with the soil | lacks a nucleus |

**2** Active transport, diffusion and osmosis are described below.

1 the movement of ions or molecules across the cell membrane against a concentration gradient using energy

2 the movement of ions or molecules from a region of high concentration to a region of low concentration down a concentration gradient

3 the movement of water molecules from a region of their higher concentration to a region of their lower concentration through a partially permeable membrane

What links the descriptions with their names?

| | diffusion | osmosis | active transport |
|---|---|---|---|
| **A** | 1 | 3 | 2 |
| **B** | 2 | 1 | 3 |
| **C** | 2 | 3 | 1 |
| **D** | 3 | 2 | 1 |

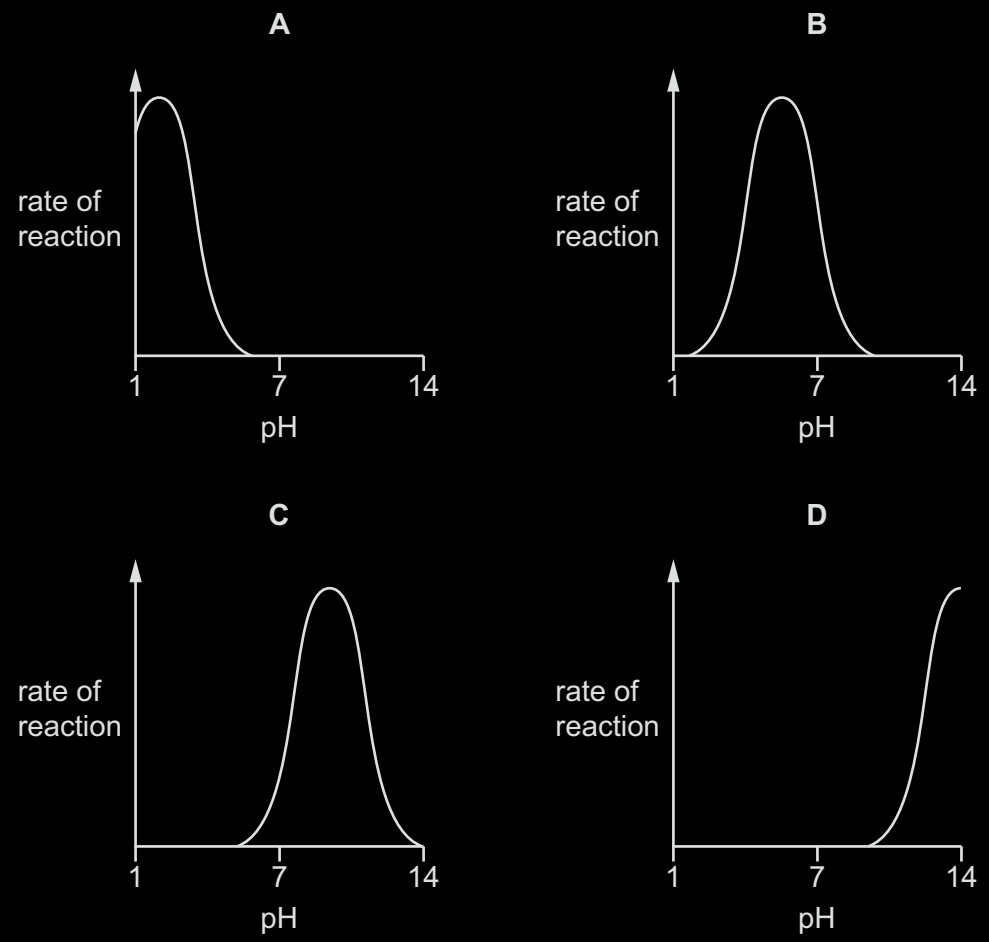
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**3** A cube of fresh potato is weighed. It is then placed in a test-tube containing a dilute solution of sucrose. After an hour, its mass has increased.

Which process has occurred and what has happened to the concentration of the sucrose in the solution in the test-tube?

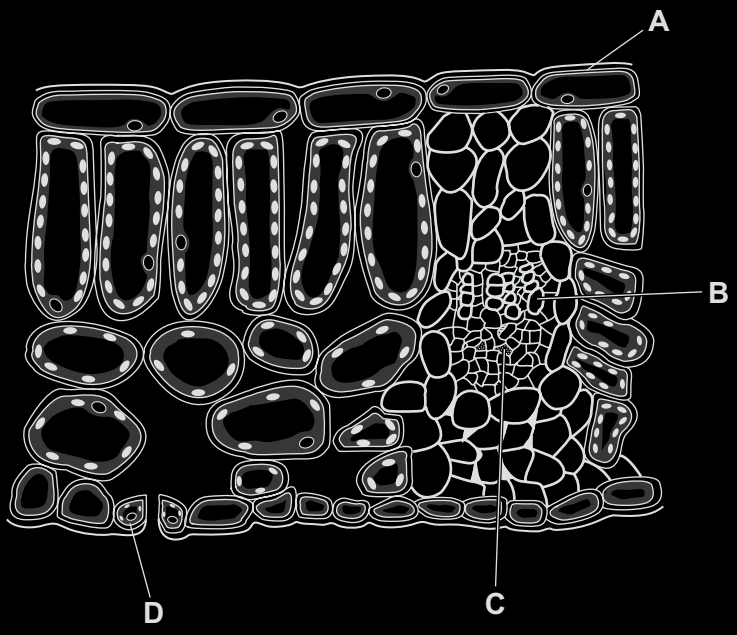
| | process | sucrose concentration |
|---|---|---|
| **A** | active transport | decreased |
| **B** | active transport | increased |
| **C** | osmosis | decreased |
| **D** | osmosis | increased |

**4** Which graph best represents the effect of pH on the action of gastric protease?



**5** The diagram shows a section through part of a dicotyledonous leaf, as seen under the microscope.

Where is the xylem?

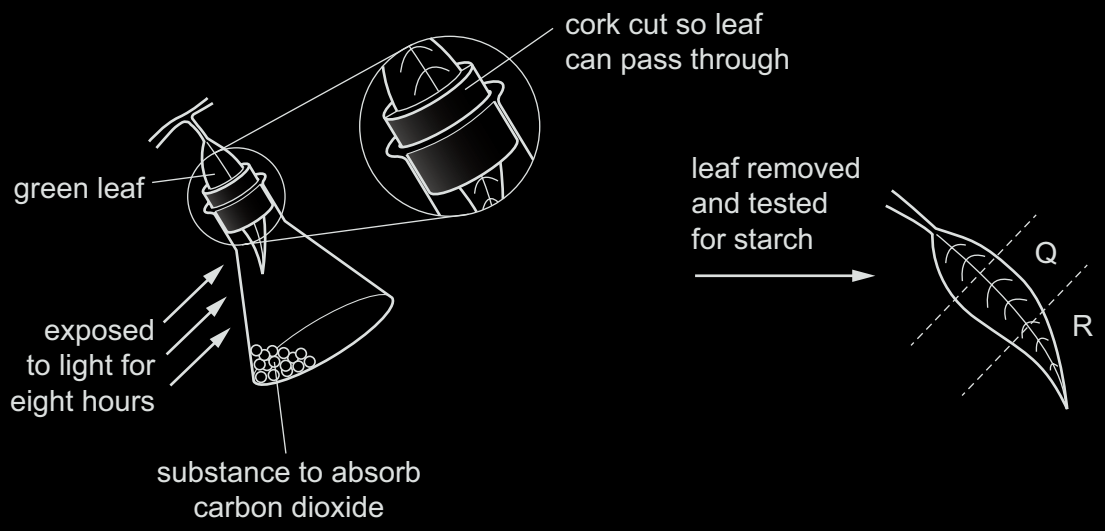



**6** When a plant is photosynthesising, which gases enter and leave the stomata?

| | carbon dioxide | oxygen | water vapour |
|---|---|---|---|
| **A** | enters | leaves | enters |
| **B** | enters | leaves | leaves |
| **C** | leaves | enters | enters |
| **D** | leaves | enters | leaves |

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**7** A plant is kept in the dark for two days. One of its leaves is used in an experiment to investigate photosynthesis as shown in the diagram.




What are the colours of Q and R, when the leaf is tested for starch using iodine solution?

| | Q | R |
|---|---|---|
| **A** | blue / black | brown |
| **B** | brown | brown |
| **C** | blue / black | blue / black |
| **D** | brown | blue / black |

**8** Which molecule is absorbed in the ileum?

**A** glycerol

**B** glycogen

**C** protein

**D** starch

**9** Which table describes the elements present in carbohydrate, protein and fat?

| | | elements | | | |
|---|---|---|---|---|---|
| | | C | N | O | H |
| **A** | carbohydrate | ✓ | ✓ | ✓ | ✓ |
| | protein | ✓ | ✗ | ✓ | ✓ |
| | fat | ✓ | ✓ | ✗ | ✓ |

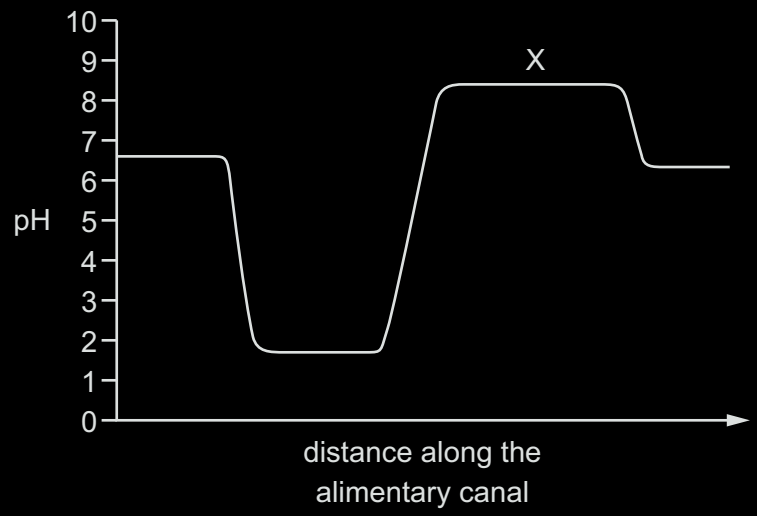
| | | elements | | | |
|---|---|---|---|---|---|
| | | C | N | O | H |
| **B** | carbohydrate | ✓ | ✗ | ✓ | ✓ |
| | protein | ✓ | ✓ | ✓ | ✓ |
| | fat | ✓ | ✗ | ✓ | ✓ |

| | | elements | | | |
|---|---|---|---|---|---|
| | | C | N | O | H |
| **C** | carbohydrate | ✓ | ✗ | ✗ | ✓ |
| | protein | ✓ | ✓ | ✓ | ✓ |
| | fat | ✓ | ✓ | ✗ | ✓ |

| | | elements | | | |
|---|---|---|---|---|---|
| | | C | N | O | H |
| **D** | carbohydrate | ✓ | ✗ | ✓ | ✓ |
| | protein | ✓ | ✓ | ✗ | ✓ |
| | fat | ✓ | ✗ | ✗ | ✓ |

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**10** The graph shows how the pH changes along the alimentary canal.




What is the region labelled X?

**A** large intestine

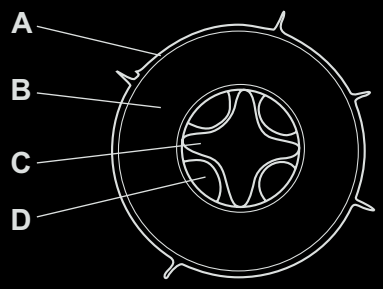
**B** oesophagus

**C** small intestine

**D** stomach

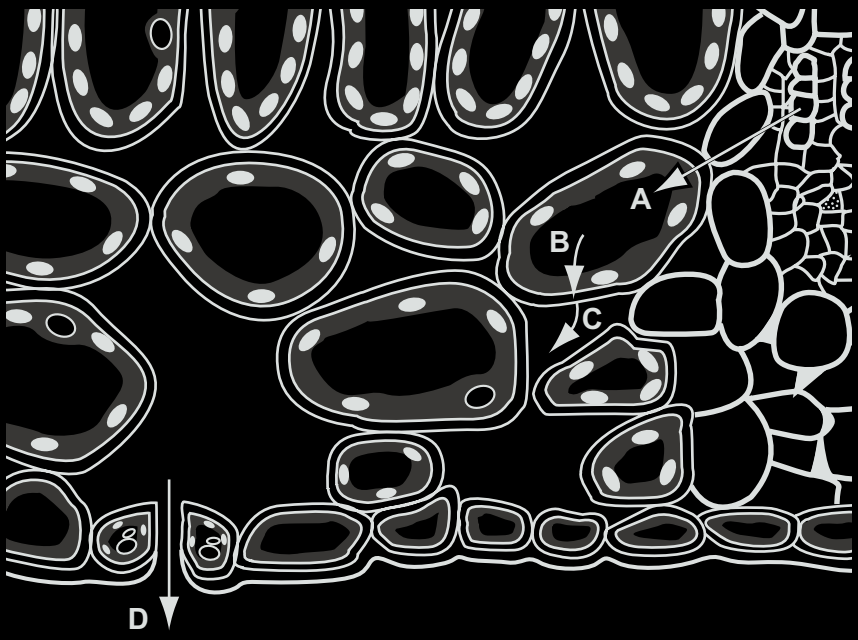
**11** The diagram shows a section through a root.

Which tissue transports amino acids?



**12** The diagram shows the movement of water molecules in a leaf from the xylem to the atmosphere.

At which stage does evaporation occur?




**13** Which organs receive blood from both an artery **and** a vein?

| | kidney | liver | lung |
|---|---|---|---|
| **A** | ✓ | ✓ | ✗ |
| **B** | ✓ | ✗ | ✓ |
| **C** | ✗ | ✓ | ✗ |
| **D** | ✗ | ✗ | ✓ |

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**14** The diagram shows a section through the human heart.



Why is the wall of the right ventricle thicker than the wall of the right atrium?

**A** The chambers have different oxygen concentrations.

**B** The right atrium must pump blood all the way around the body.

**C** The right ventricle must exert a greater force than the right atrium.

**D** The right ventricle receives more blood than the right atrium.

**15** The table shows the characteristics of the blood in one blood vessel in the body.

| oxygen concentration | carbon dioxide concentration | pressure |
|---|---|---|
| high | low | high |

Which blood vessel contains blood with these characteristics?

**A** aorta

**B** pulmonary artery

**C** pulmonary vein

**D** vena cava

**16** When breathing out forcefully, which muscles are contracting?

| | diaphragm | external intercostal muscles | internal intercostal muscles |
|---|---|---|---|
| **A** | ✓ | ✗ | ✓ |
| **B** | ✓ | ✗ | ✗ |
| **C** | ✗ | ✓ | ✗ |
| **D** | ✗ | ✗ | ✓ |

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**[Turn over**

**17** Scientists believe that absorption of mineral ions in plants requires energy from respiration.

Which observation best supports this idea?

**A** Carbohydrate is stored in the roots.

**B** Living roots give off carbon dioxide.

**C** The root hairs have a large surface area.

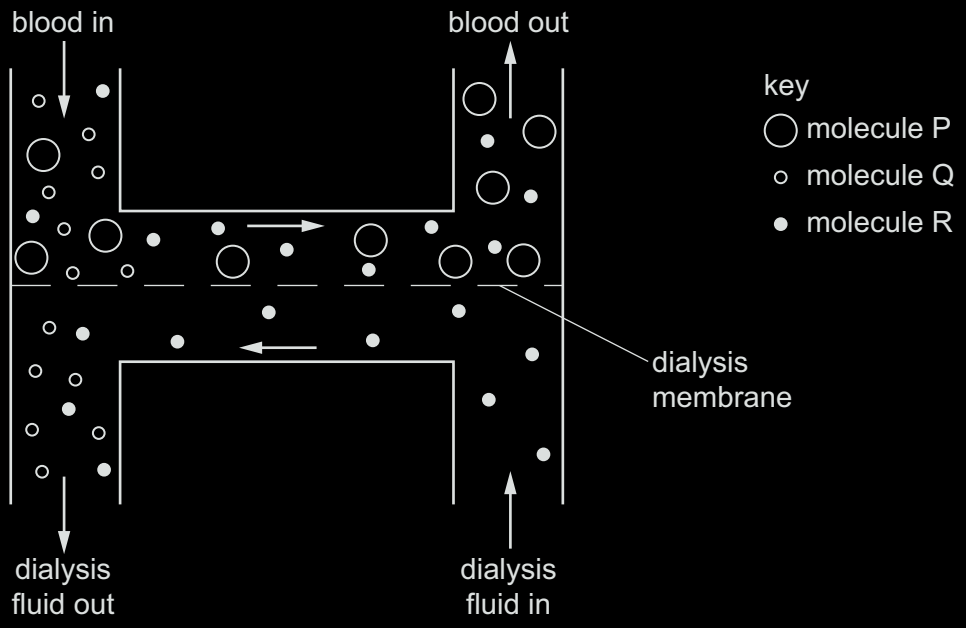
**D** Uptake of nitrate is reduced in lower oxygen concentrations.

**18** At the end of a 100 m sprint, how will blood passing through the leg muscles be different from the blood at the start of the race?

| | carbon dioxide concentration | lactic acid concentration | temperature |
|---|---|---|---|
| **A** | higher | higher | higher |
| **B** | higher | lower | lower |
| **C** | lower | higher | lower |
| **D** | lower | lower | higher |

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**19** The diagram shows what happens to molecules of glucose, protein and urea as blood passes through a kidney dialysis machine.




What are molecules P, Q and R?

| | molecule P | molecule Q | molecule R |
|---|---|---|---|
| **A** | glucose | protein | urea |
| **B** | glucose | urea | protein |
| **C** | protein | glucose | urea |
| **D** | protein | urea | glucose |

**20** Which process is responsible for maintaining a person's body temperature when they are too cold?

**A** excretion

**B** homeostasis

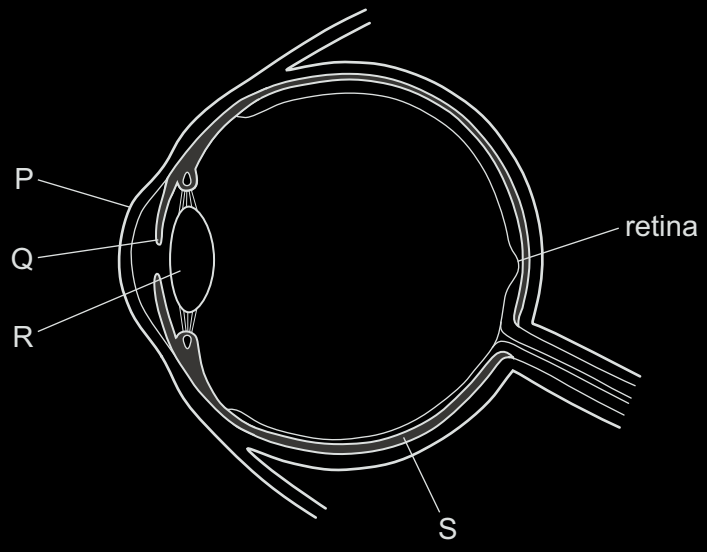
**C** sweating

**D** vasodilation

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**[Turn over**

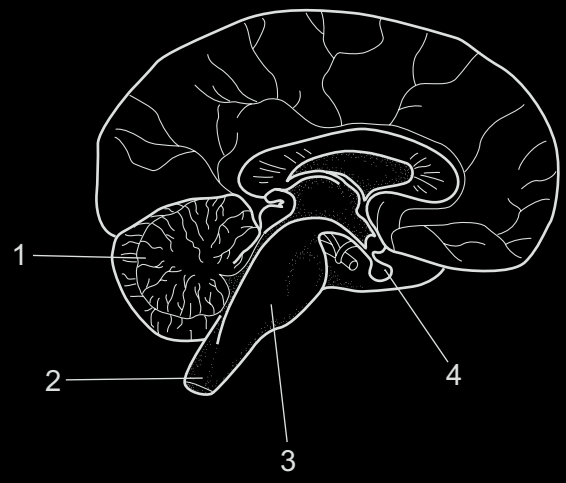
**21** The diagram shows a section through the eye.




Which pair of structures focus light rays onto the retina?

**A** P and Q **B** P and R **C** Q and R **D** Q and S

**22** The diagram represents a section through the brain and associated structures.



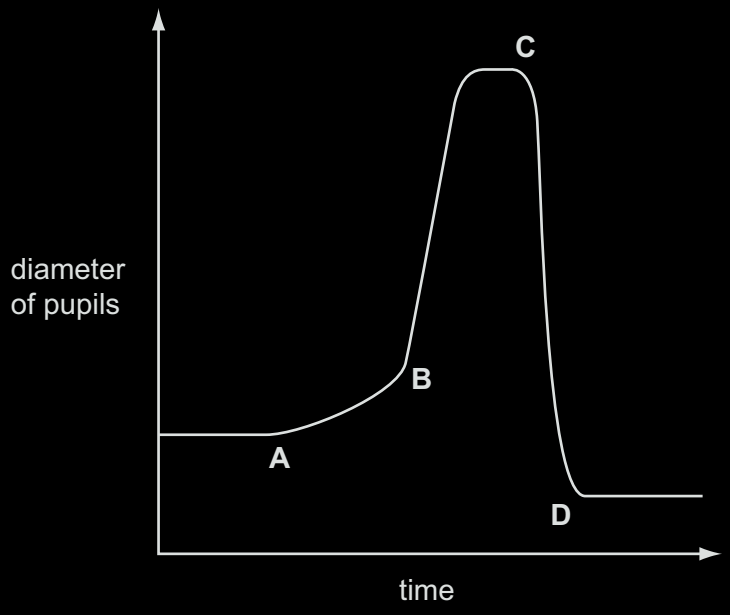

Which labelled region coordinates breathing movements and which secretes a hormone causing ovulation?

| | coordinates breathing movements | secretes a hormone causing ovulation |
|---|---|---|
| **A** | 1 | 2 |
| **B** | 1 | 4 |
| **C** | 3 | 2 |
| **D** | 3 | 4 |

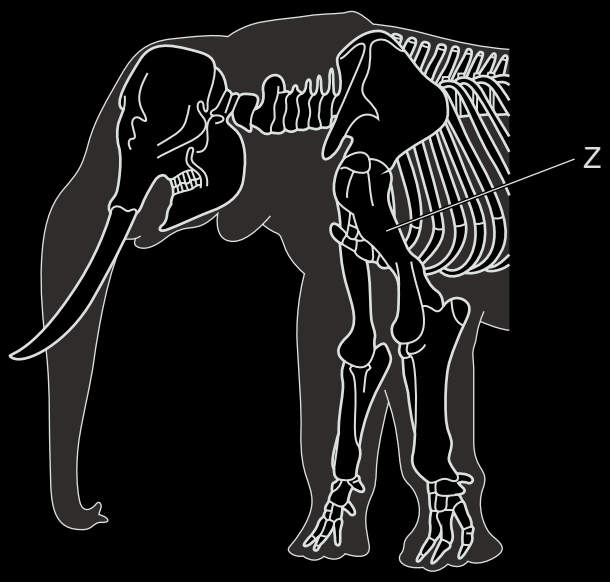
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**23** The graph shows changes in the diameter of a person's pupils while outdoors on a sunny day.

At which time did the person take off a pair of sunglasses?




**24** The diagram shows part of the skeleton of an elephant.




What is the bone labelled Z?

**A** humerus **B** radius **C** scapula **D** ulna

**25** Which risk is greatly increased by the nicotine in cigarette smoke?

**A** bronchitis and emphysema

**B** cancer of the airways

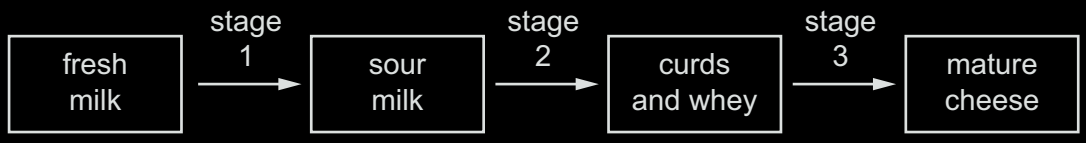
**C** high blood pressure

**D** reduced birth weight of babies

**26** Which type of organism produces penicillin, and which type of infection can be treated using penicillin?

| | type of organism | type of infection treated |
|---|---|---|
| **A** | bacterium | bacterial |
| **B** | bacterium | viral |
| **C** | fungus | bacterial |
| **D** | fungus | viral |

**27** The diagram shows three stages in the production of a blue cheese.




The three stages use micro-organisms and enzymes, as shown by P, Q, R and S.

P bacterial culture added

Q bacteria killed by pasteurisation

R rennin added

S fungal mould added

What is done at each of the stages?

| | stage 1 | stage 2 | stage 3 |
|---|---|---|---|
| **A** | P and then Q | R | S |
| **B** | P and then Q | S | R |
| **C** | Q and then P | R | S |
| **D** | Q and then P | S | R |

**28** Why does an ecosystem need to be exposed regularly to sunlight?

**A** Energy is converted to biomass.

**B** Energy is lost as heat.

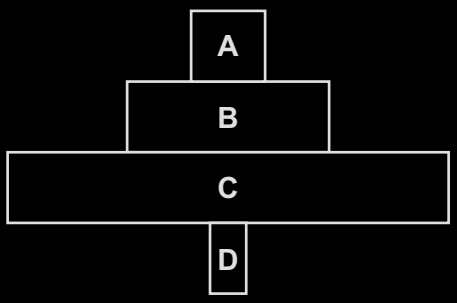
**C** Energy is lost to decomposers.

**D** Energy is reflected by plants.

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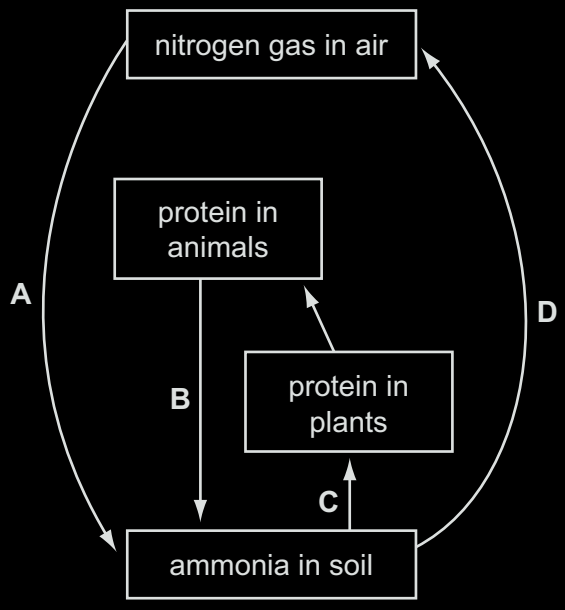
**29** The diagram shows a pyramid of numbers in a woodland ecosystem.

At which trophic level are the individual organisms **largest in size**?




**30** The diagram shows part of the nitrogen cycle.

Which arrow shows decomposition?




**31** Three statements about malarial parasites are listed.

1 Insecticides are used to kill the vectors.
2 Netting is used to keep the vectors away from people.
3 People take drugs that stop the malarial pathogen developing.

Which methods can be used to control malaria?

**A** 1, 2 and 3

**B** 1 and 2 only

**C** 1 only

**D** 2 and 3 only

**32** Which consequence of burning fossil fuel leads to acid rain?

**A** Burning carbon compounds produces carbon dioxide.

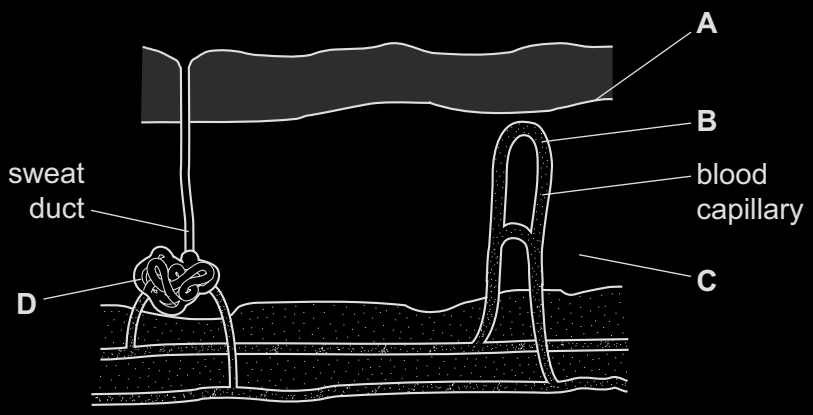
**B** Burning carbon compounds produces carbon monoxide.

**C** Burning sulfur compounds produces sulfur dioxide.

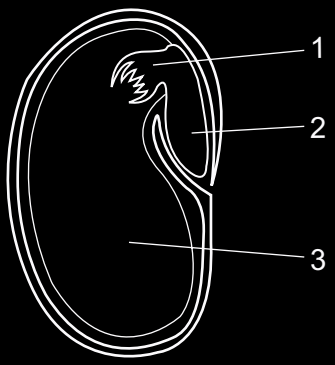
**D** Burning sulfur compounds produces water vapour.

**33** The diagram represents a section through the human skin.

In which part is mitosis occurring most rapidly?




**34** The diagram shows a section of a seed.




What are the numbered parts?

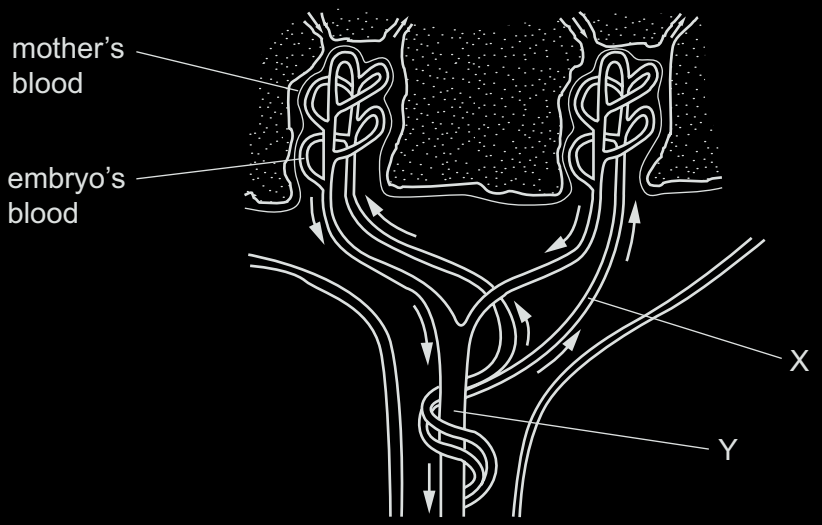
| | 1 | 2 | 3 |
|---|---|---|---|
| **A** | cotyledon | plumule | radicle |
| **B** | plumule | cotyledon | radicle |
| **C** | plumule | radicle | cotyledon |
| **D** | radicle | plumule | cotyledon |

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**35** What are the functions of LH and FSH in the menstrual cycle?

| | LH | FSH |
|---|---|---|
| **A** | causes ovulation | causes menstruation |
| **B** | causes ovulation | stimulates an egg to mature in an ovary |
| **C** | stimulates an egg to mature in an ovary | causes menstruation |
| **D** | stimulates an egg to mature in an ovary | causes ovulation |

**36** The diagram shows how the blood of a human embryo flows close to the mother's blood in the placenta.




Which substances are present at X in higher concentrations than at Y?

**A** carbon dioxide and glucose

**B** carbon dioxide and urea

**C** glucose and oxygen

**D** glucose and urea

**37** Which term describes reproduction between those members of a species that are best fitted to their environment?

**A** discontinuous variation

**B** gene mutation

**C** natural selection

**D** survival of the fittest

**38** In the ABO blood grouping system, which genotype is homozygous dominant?

**A** $I^A\ I^O$ **B** $I^A\ I^B$ **C** $I^B\ I^B$ **D** $I^O\ I^O$

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**39** Bacteria can be genetically modified to produce human insulin.

What is a possible danger of this procedure?

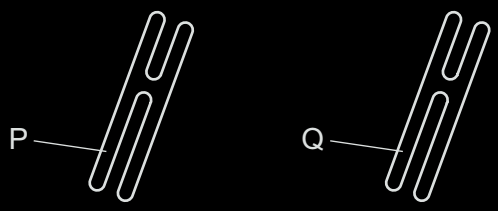
**A** Bacterial insulin is less effective in treating diabetes than animal insulin.

**B** The genetically modified bacteria may become insulin resistant.

**C** The genetically modified bacteria may produce too much insulin.

**D** The presence of a new gene in the bacteria may alter the way that existing genes work.

**40** The diagram shows a pair of chromosomes from the same cell.




A gene is found at the point labelled P.

In a heterozygous individual, what will be found at the equivalent position labelled Q?

**A** a different allele of a different gene

**B** a different allele of the same gene

**C** a different gene of the same allele

**D** the same gene of the same allele

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